



CHAPTER 3

INTRODUCING DEVOPS





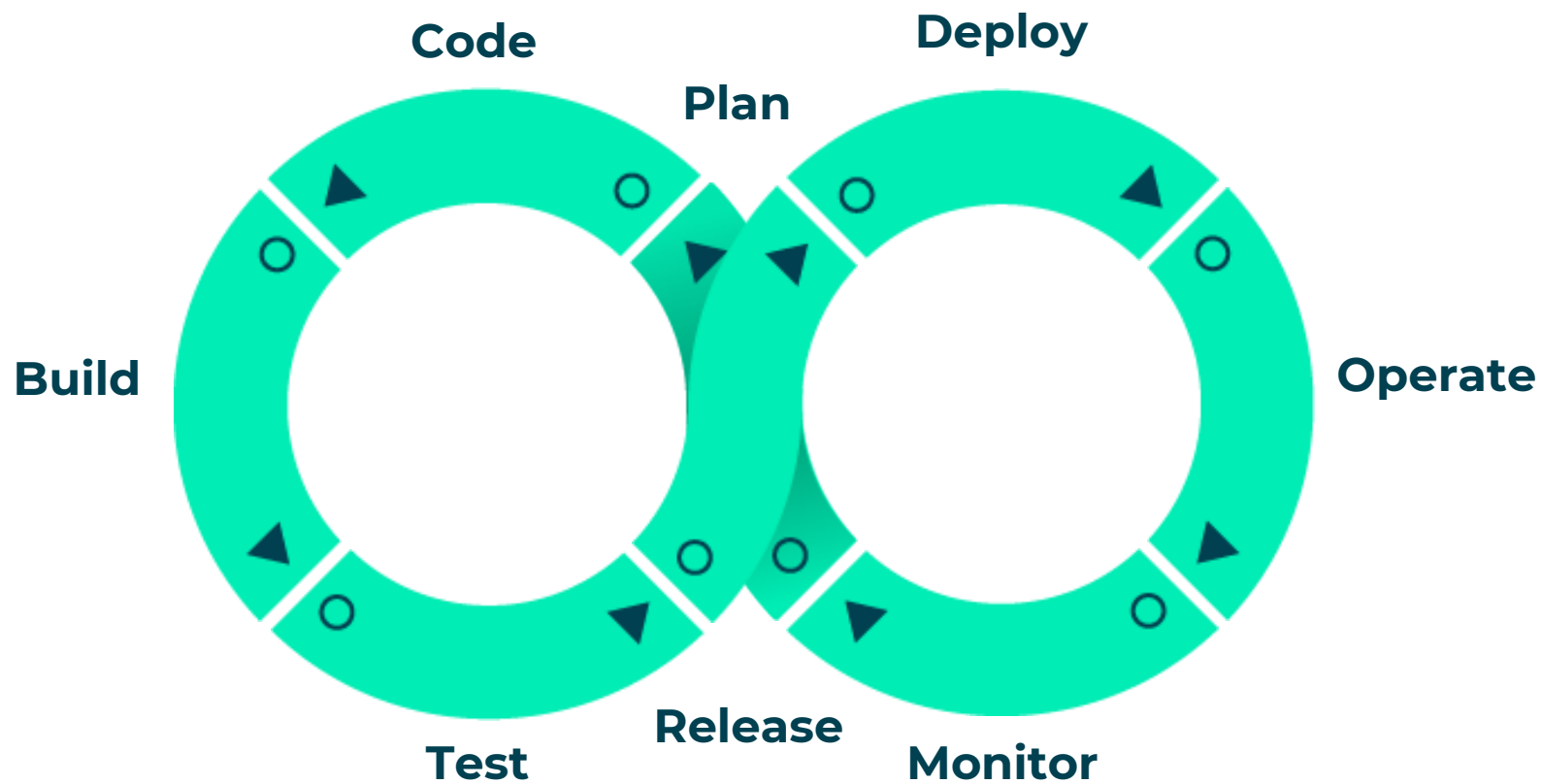
TOPICS

Understanding DevOps

Building a DevOps culture

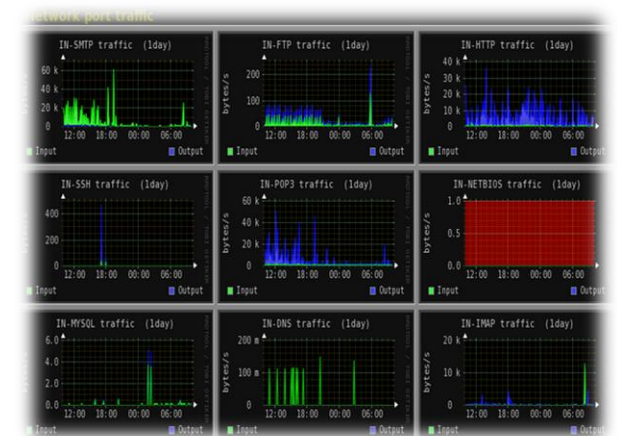


Software Development never ends...



QA Common Problems DevOps Attempts to Solve

- **Request for More Server Capacity:**
 - We need to increase our server capacity to handle the growing demand.
- **Testing in Production:**
 - “Wouldn’t it be great if we could test our system directly in the production environment?”
- **Accountability for Issues:**
 - An issue has occurred. Who is responsible?
- **Monitoring System Alerts:**
 - We get meaningless alerts!





I NEED MORE SERVER(NODE) CAPACITY

- **The most common complaint from the business**
 - Ops team are seen as blockers who are not sensitive to the needs of business
- **Capacity demand often fluctuates without clear reasons**
- **Ops teams manage more than just your app**
 - 🚨 System outages
 - 🐛 System defects
 - 🛡️ Incident and Security Management
 - 💰 Resource and Budget Management
 - 📊 Monitoring and Improving KPIs
 - 📄 Compliance



ANOTHER PROBLEM – WHO CAUSED IT THIS TIME?



**Production
environment**

Who should we
take aim at?



Dev Team

Dev Team: Why can't
build software that
works?

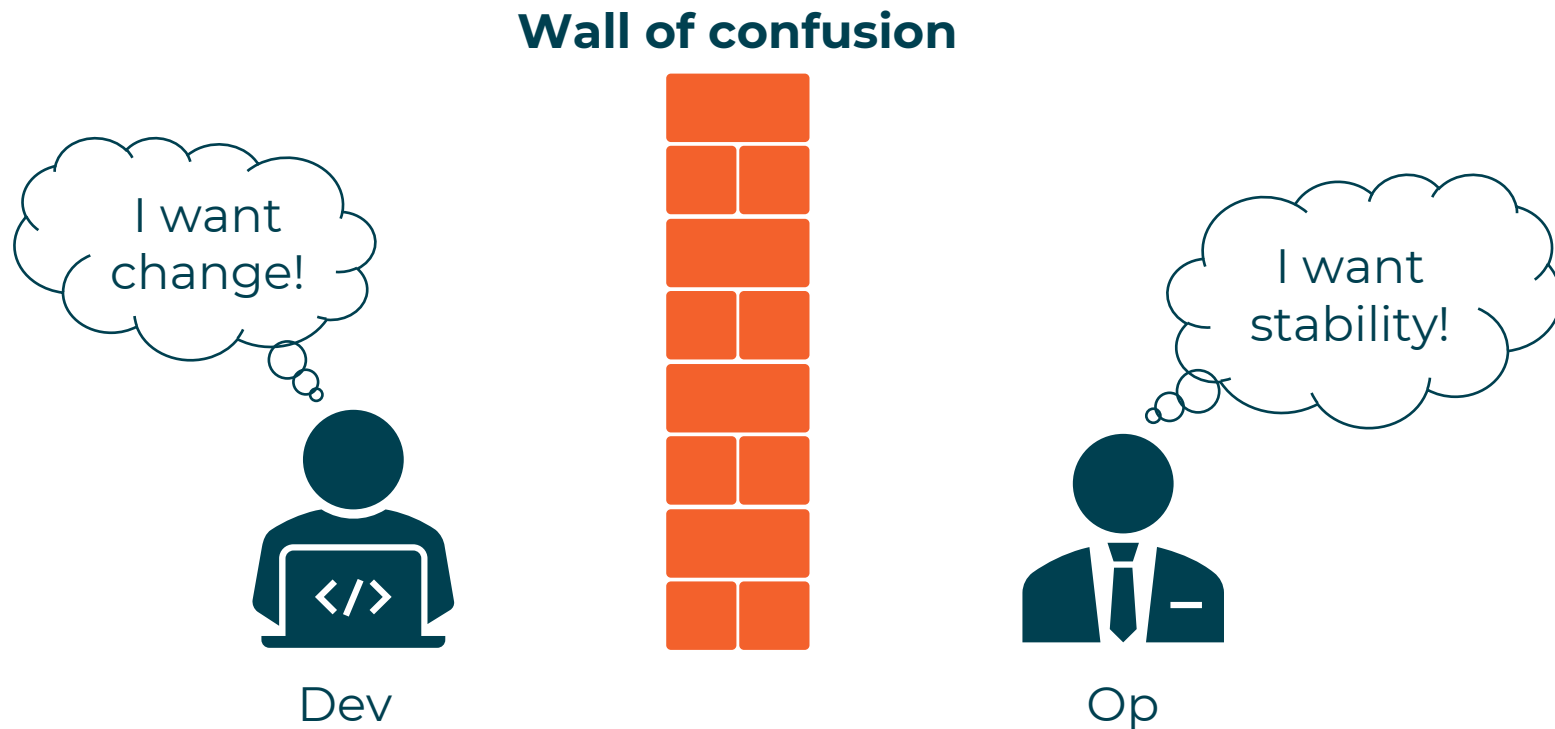


Business

Ops Team: Why can't
you get the capacity
right?



THE WALL OF CONFUSION

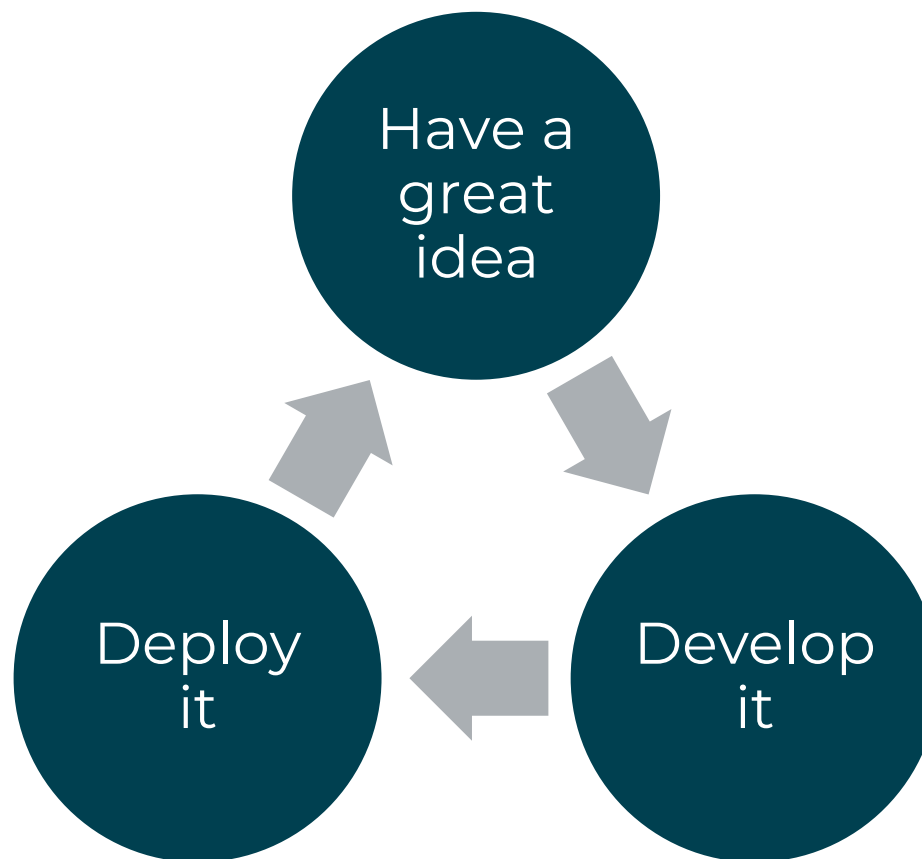


**What about Security, Governance,
Risk Management and Compliance?**

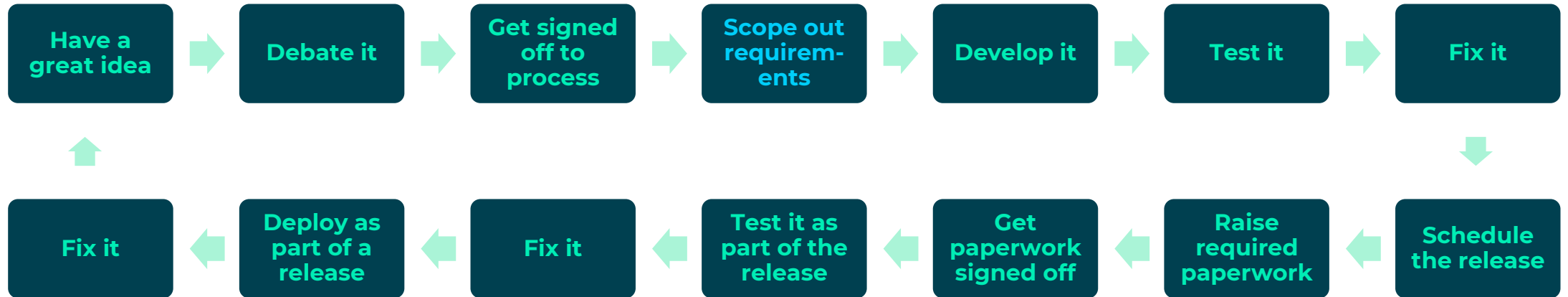
**OPs wants to maintain stability, availability
Dev wants to create change - Conflicting views**



START-UPS – GREEN FIELDS

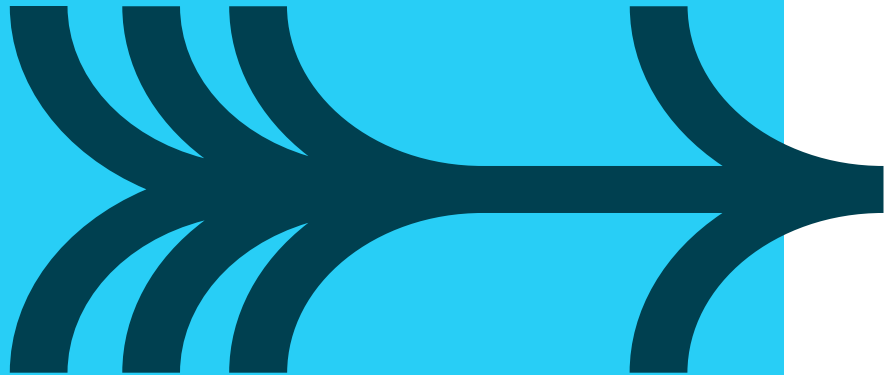


QA Enterprise – Complexity Due to Scale





DEVOPS AS A NEW ROLE?



Devs perspective

CI/CD

First Deployment

Monitored

Tested

Ops perspectives

Virtualisation

Cloud based

High availability

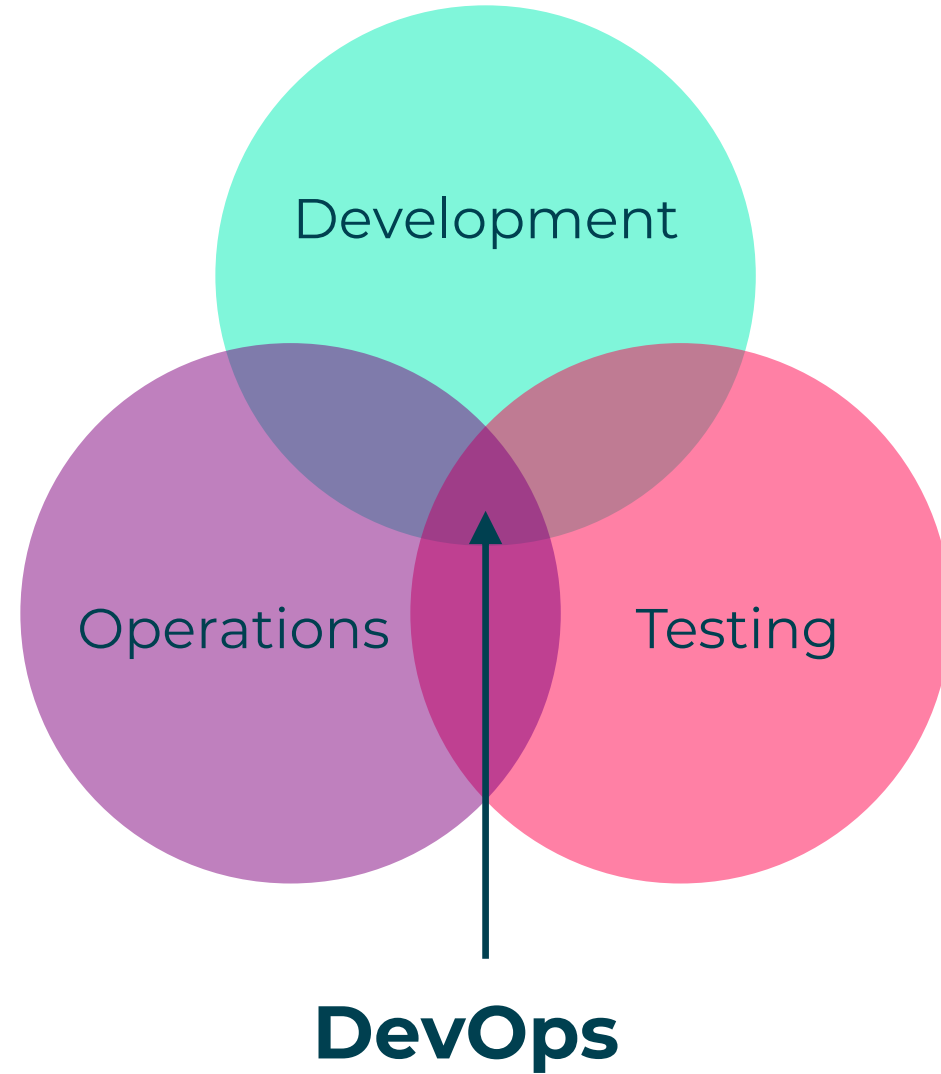
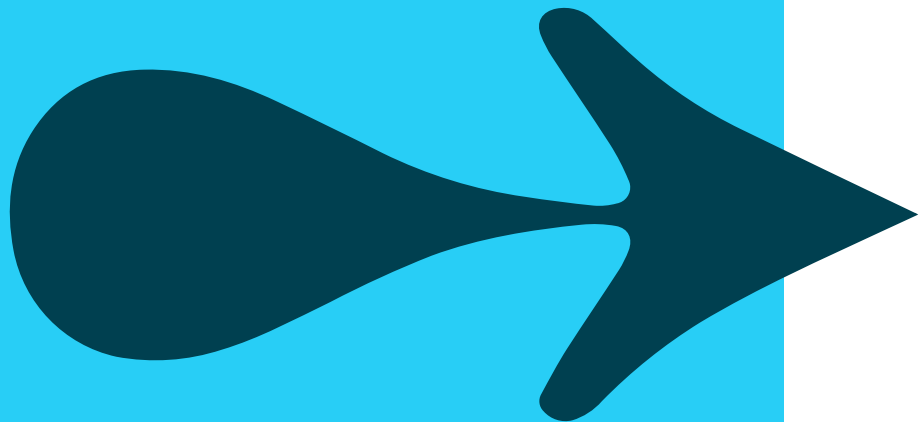
Testable

Maintainable

Reproducible



DEVOPS – WE'RE
ALL IN THIS
TOGETHER!





KEEP CALMS AND DO DEVOPS



DevOps – The C.A.L.M.S Lifecycle:

C – Culture

A – Automation

L – Lean

M – Measurement

S – Sharing



CULTURE

 **SHARING THE SAME VISION, GOALS AND INCENTIVES**

 **OPEN, HONEST, TWO-WAY COMMUNICATION**

 **COLLABORATION**

 **PRIDE OF WORK**

 **RESPECT OTHERS**

 **TRUST OTHERS**

 **TRANSPARENCY**





Change of culture

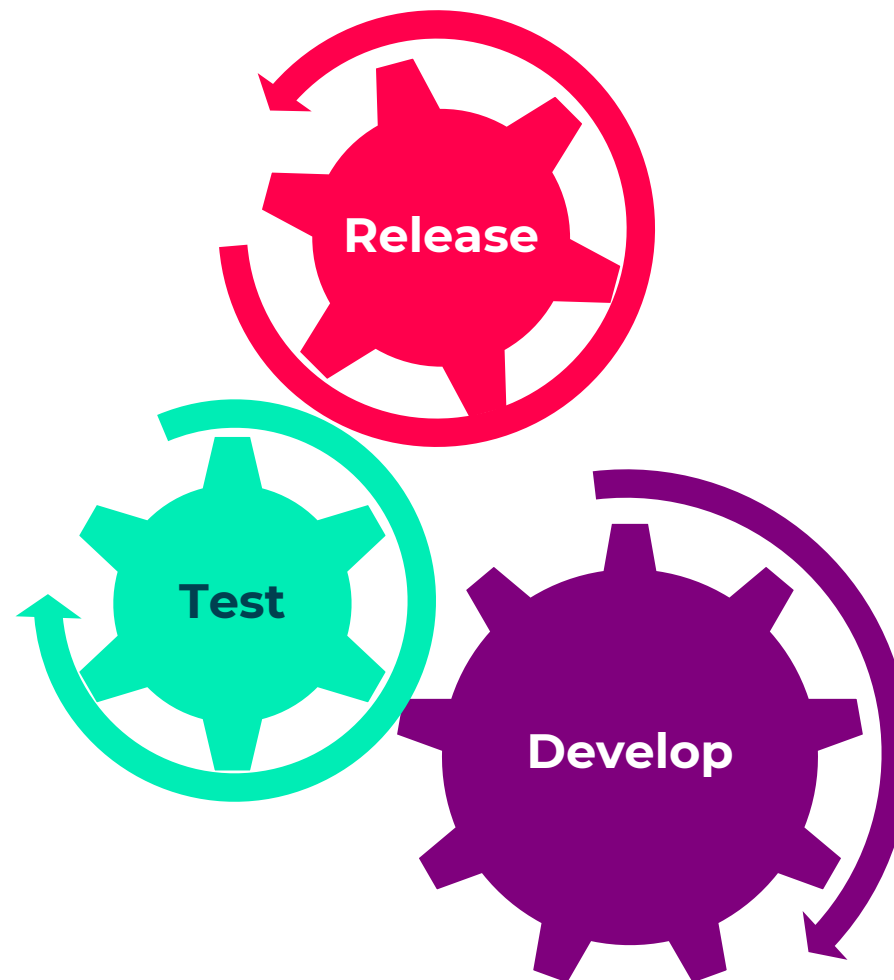
From	To
 IT only thinking	Customer focus
 Teams stuck in silos	Cross-functional teams
 Control over collaboration	Collaborative
 Tasks over outcomes	Outcome-oriented
 Blame instead of learning	Taking on responsibility
 Always firefighting	Proactive
 Comfort in the past	Courageous
 Change is resisted	Flexible
 Trust is scarce	High trust
 Mistakes are hidden	Share responsibilities, report failure
 New ideas do not survive	Novelty is implemented

People don't resist change; they resist being changed. Peter Senge



A – AUTOMATION

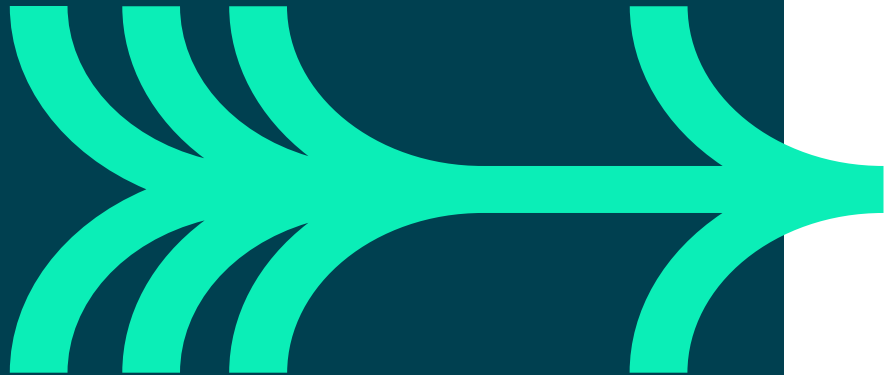
Tools make automation possible



**Amazon push code to production
every 11.7 seconds**



WE NEED ROUNDED PROFESSIONALS



The agile world demands multifaceted teams

- Dev teams should be involved end-to-end





L – LEAN



Eliminate Waste



Build Quality In



Create Knowledge



Defer Commitment



Deliver Fast



Respect People



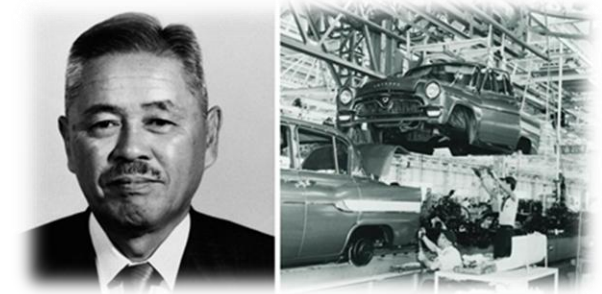
Optimise The Whole



CONTINUOUS IMPROVEMENT

Continuous improvement

- **Kaizen** – 'good change' – continuous improvement mindset



Taiichi Ohno – the father of Lean

Targeting areas for improvement

- Bottlenecks
- Waste elimination
- Reduction of variability





We Use Tools for:



Automation



Orchestration



Configuration



Testing



Logging and Monitoring

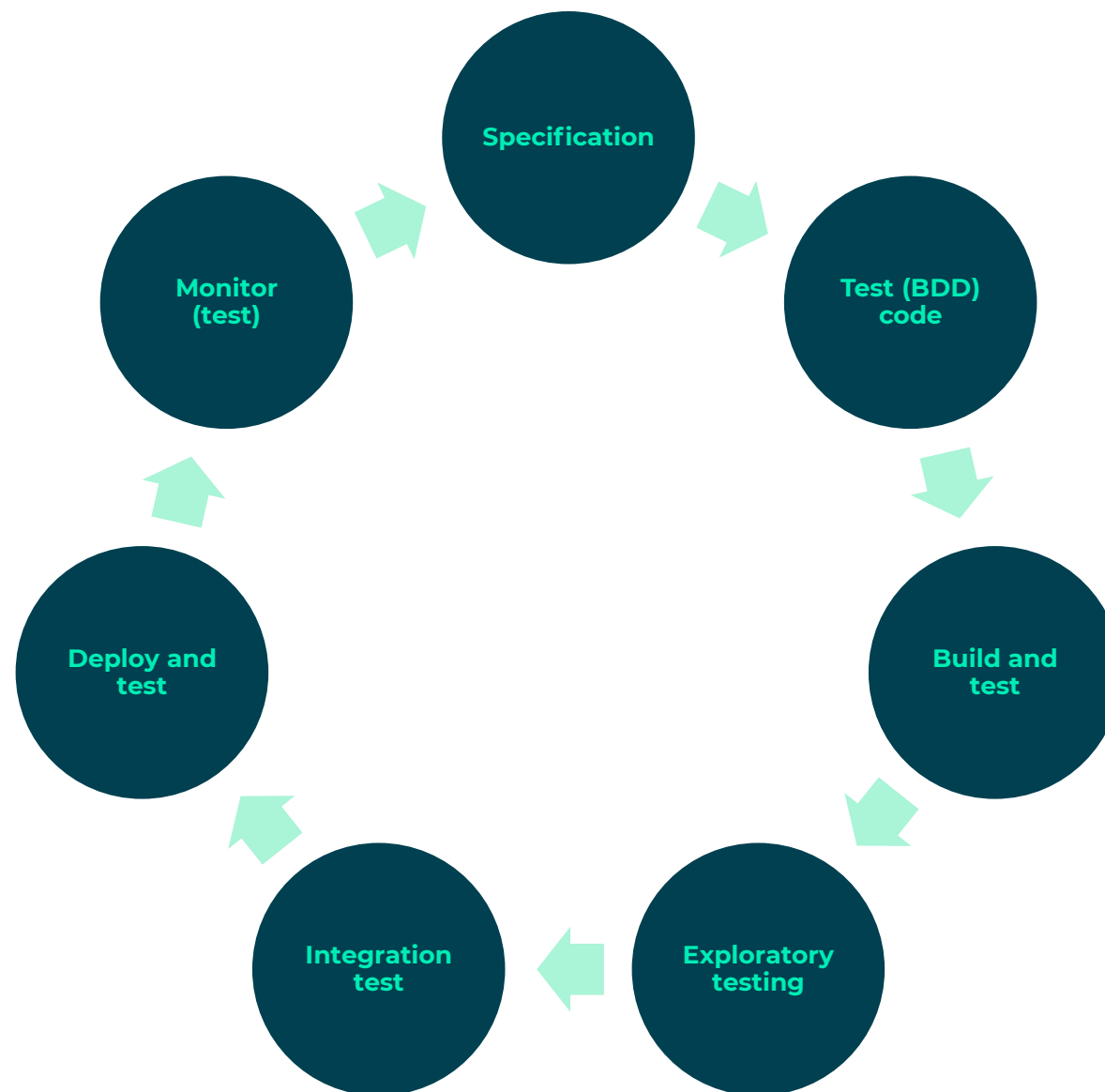


CI CD





TEST BASED DEVOPS





THANK YOU

Hope you enjoyed this learning journey.





WHAT IS NOOPS?

NoOps stands for no operations

- IT environment can be so automated and abstracted that no dedicated in-house team is needed to manage software
- Main drivers are IT automation and cloud computing





WHAT IS WEBOPS?



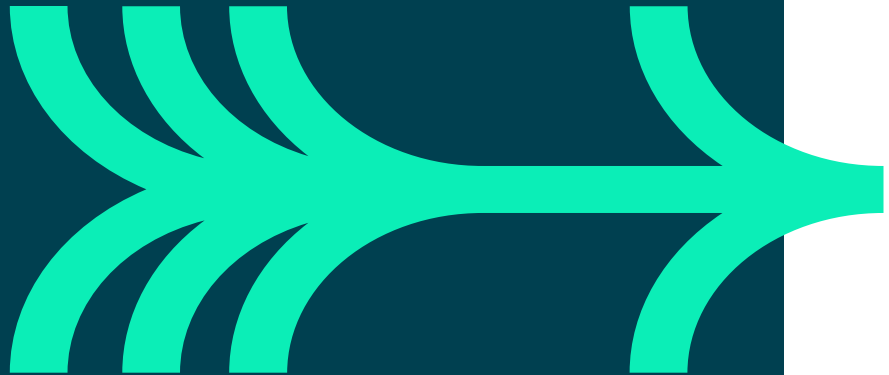
WebOps, short for Web Operations

- Focuses on the development, deployment, maintenance, and scaling of web applications. It enhances efficiency and performance through automation and specialised web tools
- **The area of WebOps engineering includes:**
 - Application deployment
 - Management
 - Maintenance
 - Configuration
 - Repair





A LESSON FROM NETFLIX



Simian Army

- Software that reap havoc on their production systems

A new approach to architectures – microservices

- Teams must clearly understand what service failure means to the system as a whole
- Live service testing is becoming a new norm – Netflix simian army used to test resilience and monitoring
- A lot of real-time monitoring
- Detect – requests/second, and transactions/second
- Respond
- Performing “semantic monitoring”
- Run subset of automated tests against live system